

TP - Project Draft

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Intro

This project analyzes an E-Commerce Transactions dataset containing over 540,000 online retail transactions between December 2010 and December 2011. Each record includes details such as invoice number, product description, quantity purchased, unit price, transaction date, customer ID, and country. The dataset provides a rich source for examining customer purchasing behavior, product demand patterns, and overall sales trends. The goal of this project is twofold: (1) conduct a rigorous statistical analysis in R to understand the key drivers of purchase amounts and uncover business insights, and (2) design an interactive Tableau dashboard that communicates these insights in a clear, business-friendly format. By combining regression modeling with modern data visualization, this project aims to highlight how data science can guide decision-making in e-commerce, from identifying high-value customers to optimizing product strategy.

```
transactions <- read_csv("data.csv")
#summary(transactions)
```

```
head(transactions)
```

```
# A tibble: 6 x 8
  InvoiceNo StockCode Description      Quantity InvoiceDate UnitPrice CustomerID
  <chr>      <chr>      <chr>          <dbl> <chr>          <dbl>      <dbl>
1 536365    85123A    WHITE HANGING H~         6 12/1/2010 ~         2.55      17850
2 536365    71053    WHITE METAL LAN~         6 12/1/2010 ~         3.39      17850
3 536365    84406B    CREAM CUPID HEA~         8 12/1/2010 ~         2.75      17850
4 536365    84029G    KNITTED UNION F~         6 12/1/2010 ~         3.39      17850
5 536365    84029E    RED WOOLLY HOTT~         6 12/1/2010 ~         3.39      17850
6 536365    22752    SET 7 BABUSHKA ~         2 12/1/2010 ~         7.65      17850
# i 1 more variable: Country <chr>
```

Initial Data Summary and Observations

The dataset contains 541,909 transaction records and 8 variables: `InvoiceNo`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `UnitPrice`, `CustomerID`, and `Country`. A preliminary summary reveals several important characteristics and data quality issues:

- **InvoiceNo, StockCode, Description, and Country** are stored as text fields, which is expected since they represent transaction identifiers, product descriptions, or categorical values.
- **Quantity** ranges from **-80,995 to 80,995**. Negative values represent product returns or cancellations, while most transactions involve small positive quantities (median = 3, 75th percentile = 10).
- **InvoiceDate** is currently stored as a character variable and requires conversion to a proper datetime format for time-based analysis.
- **UnitPrice** ranges from **-11,062 to 38,970**. Negative or zero prices likely indicate errors or adjustments, whereas the majority of products are low-priced items (median = 2.1, mean = 4.6).
- **CustomerID** ranges from 12,346 to 18,287 but has **135,080 missing values**, representing about **25% of transactions**. These may correspond to guest checkouts or incomplete records.

Overall, the dataset is rich in detail but requires substantial cleaning before statistical analysis. Specifically, returns/cancellations, extreme outliers, and missing customer identifiers must be addressed to ensure reliable modeling and dashboard insights.

###Data Cleaning

```
transactions_clean <- transactions %>%
  # Step 1: Convert InvoiceDate
  mutate(InvoiceDate = mdy_hm(InvoiceDate)) %>%

  # Step 2: Handle negative quantities (create flag, filter extremes)
  mutate(ReturnFlag = Quantity < 0) %>%
  filter(Quantity > -10000 & Quantity < 10000) %>%

  # Step 3: Handle UnitPrice anomalies
  mutate(PriceFlag = UnitPrice <= 0) %>%
  filter(UnitPrice > 0 & UnitPrice < 10000) %>%

  # Step 4: Handle missing CustomerID
  mutate(
    CustomerID = as.character(CustomerID),
    CustomerID = ifelse(is.na(CustomerID) | CustomerID == "" | CustomerID == "NA",
```

```

        "Guest", CustomerID)
    ) %>%

# Step 5: Standardize text fields
mutate(
  Description = str_to_title(str_trim(Description)),
  Country = str_to_title(str_trim(Country))
) %>%

# Step 6: Remove duplicate transactions
distinct(InvoiceNo, StockCode, .keep_all = TRUE) %>%

# Step 7: Derived variables
mutate(
  TotalPrice = Quantity * UnitPrice,
  TransactionMonth = month(InvoiceDate, label = TRUE),
  TransactionYear = year(InvoiceDate)
)

#head(transactions_clean )
#summary(transactions_clean )
write.csv(transactions_clean, "clean_ecommerce.csv", row.names = FALSE)
unique(transactions_clean$Country)

```

```

[1] "United Kingdom"      "France"              "Australia"
[4] "Netherlands"        "Germany"             "Norway"
[7] "Eire"                "Switzerland"         "Spain"
[10] "Poland"              "Portugal"            "Italy"
[13] "Belgium"             "Lithuania"           "Japan"
[16] "Iceland"             "Channel Islands"     "Denmark"
[19] "Cyprus"              "Sweden"              "Austria"
[22] "Israel"              "Finland"             "Bahrain"
[25] "Greece"              "Hong Kong"           "Singapore"
[28] "Lebanon"             "United Arab Emirates" "Saudi Arabia"
[31] "Czech Republic"     "Canada"              "Unspecified"
[34] "Brazil"              "Usa"                 "European Community"
[37] "Malta"               "Rsa"

```

```
readLines("clean_ecommerce.csv", n = 5)
```

```
[1] "\"InvoiceNo\\",\"StockCode\\",\"Description\\",\"Quantity\\",\"InvoiceDate\\",\"UnitPrice\\",\"
```

```
[2] "\"536365\\",\"85123A\\",\"White Hanging Heart T-Light Holder\\",6,2010-12-01 08:26:00,2.55
[3] "\"536365\\",\"71053\\",\"White Metal Lantern\\",6,2010-12-01 08:26:00,3.39,\"17850\\",\"Uni
[4] "\"536365\\",\"84406B\\",\"Cream Cupid Hearts Coat Hanger\\",8,2010-12-01 08:26:00,2.75,\"1
[5] "\"536365\\",\"84029G\\",\"Knitted Union Flag Hot Water Bottle\\",6,2010-12-01 08:26:00,3.3
```

Exploratory Data Analysis (EDA)

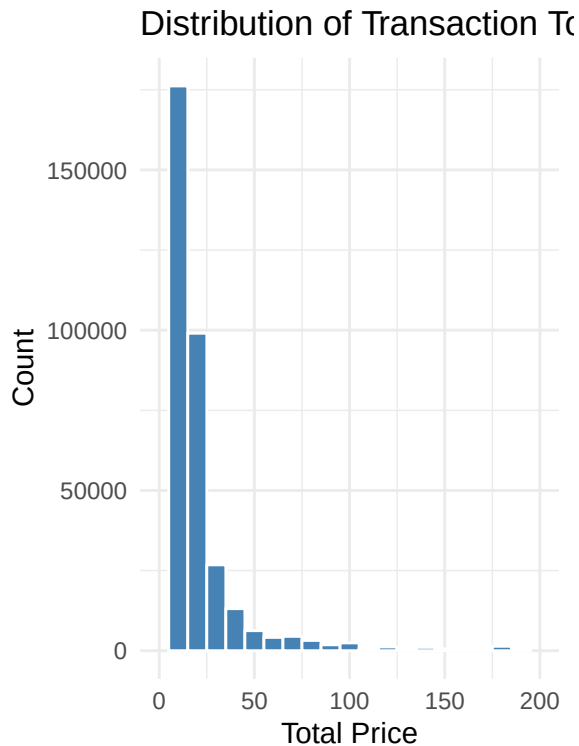
After cleaning the dataset, the next step was to conduct a thorough exploratory analysis to understand the key patterns, trends, and relationships in the data. The goal of this stage is to inform the regression modeling and identify metrics that can be highlighted in the Power BI dashboard.

1. Distribution of Transaction TotalPrice We first examined the distribution of `TotalPrice`, which represents the total value of each transaction. A histogram was generated to visualize how sales are distributed across transactions. Most transactions fall within a moderate price range, while a few extreme outliers represent unusually large purchases. Understanding this distribution helps in deciding whether transformations (e.g., log-transform) are needed for regression modeling.

```
ggplot(transactions_clean, aes(x = TotalPrice)) +
  geom_histogram(binwidth = 10, fill = "steelblue", color = "white") +
  scale_x_continuous(limits = c(0, 200)) + # focus on typical range
  labs(title = "Distribution of Transaction TotalPrice", x = "Total Price", y = "Count") +
  theme_minimal()
```

Warning: Removed 13993 rows containing non-finite values (``stat_bin()``).

Warning: Removed 2 rows containing missing values (``geom_bar()``).

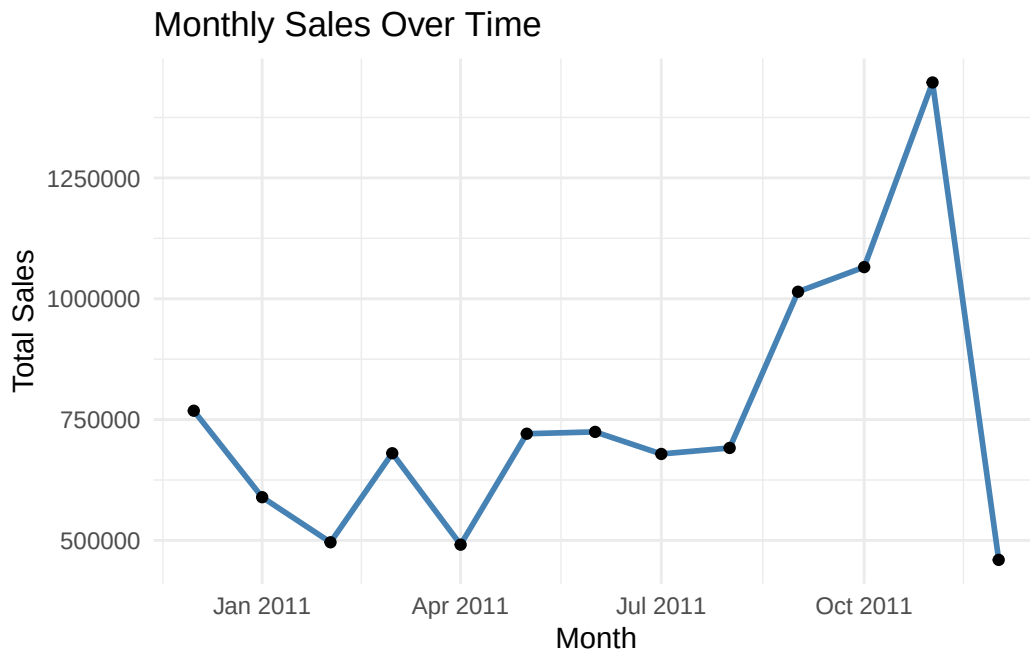


2. Monthly Sales Trends To identify seasonal patterns and sales dynamics over time, monthly sales totals were aggregated using the `InvoiceMonth` variable. A line chart was plotted to visualize total sales per month. This allows us to detect peaks and troughs in sales, which may correspond to seasonal demand, promotional campaigns, or other temporal effects.

```
sales_time <- transactions_clean %>%
  group_by(TransactionYear, TransactionMonth) %>%
  summarise(MonthlySales = sum(TotalPrice), .groups = "drop")

sales_time <- sales_time %>%
  mutate(
    MonthDate = as.Date(paste0(TransactionYear, "-", TransactionMonth, "-01"), format = "%Y-%m-%d")
  )

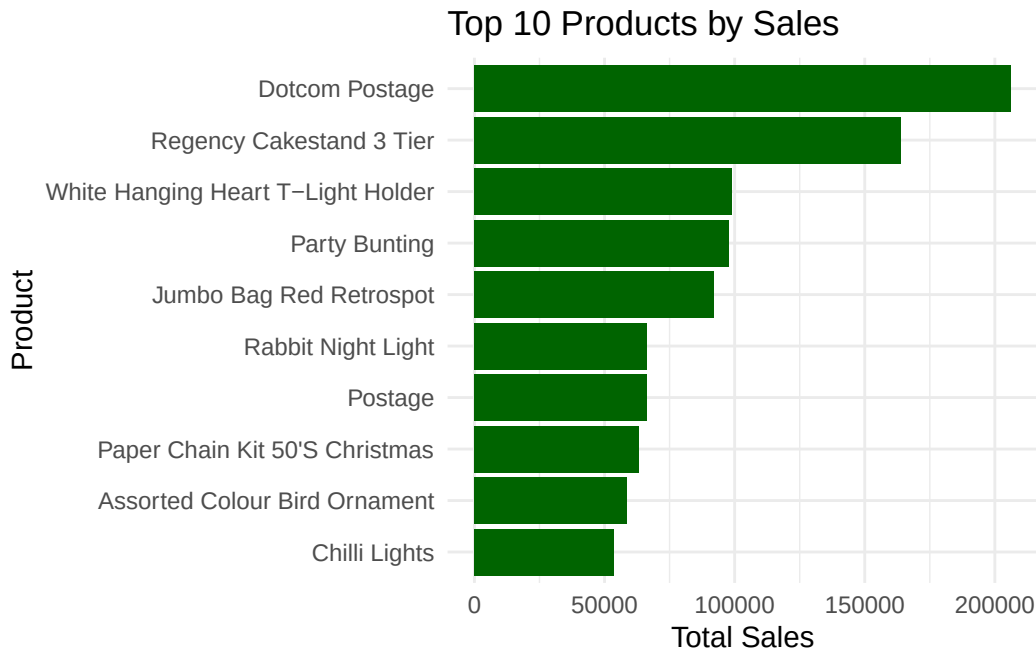
ggplot(sales_time, aes(x = MonthDate, y = MonthlySales)) +
  geom_line(color = "steelblue", linewidth = 1) + # use linewidth instead of size
  geom_point(color = "black") +
  labs(title = "Monthly Sales Over Time", x = "Month", y = "Total Sales") +
  theme_minimal()
```



3. Product Performance Next, we analyzed which products contributed most to overall revenue. By aggregating total sales per product description, we identified the top 10 best-selling products. This insight is critical for understanding product-level revenue drivers and will inform the dashboard, where highlighting top products is an important KPI.

```
top_products <- transactions_clean %>%
  group_by(Description) %>%
  summarise(TotalSales = sum(TotalPrice)) %>%
  arrange(desc(TotalSales)) %>%
  slice(1:10)

ggplot(top_products, aes(x = reorder(Description, TotalSales), y = TotalSales)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(title = "Top 10 Products by Sales", x = "Product", y = "Total Sales") +
  theme_minimal()
```



4. Customer Behavior Customer-level aggregates were also calculated to understand purchasing behavior. By summing total spend per customer, we identified the top 10 highest-spending customers. This analysis can be used to explore high-value customers and potential loyalty programs or targeted marketing opportunities.

5. Geographic Insights Sales were also aggregated by country to explore geographic trends. The top 10 countries by total sales were visualized, providing insights into key markets that may drive the majority of revenue. This is particularly useful for dashboards where stakeholders may want to analyze sales by region.

6. Quantity vs TotalPrice Relationship Finally, a scatter plot of `Quantity` versus `TotalPrice` was created to understand the relationship between the number of units purchased and total transaction value. This helps in evaluating whether larger quantities consistently generate higher revenue and may inform regression predictors.

Overall, the exploratory analysis provided a clear understanding of the dataset's structure, key trends, and business-relevant metrics. These insights will guide the regression modeling and form the foundation for the interactive dashboard, ensuring that the project highlights actionable business intelligence.